

Module 2 – Outbound Security

Module 2 Instructions – Outbound Protection

Overview

In this module, you will be configuring outbound access protection (OAP) capabilities – Managed Private endpoints (MPE) and Connection Policies.

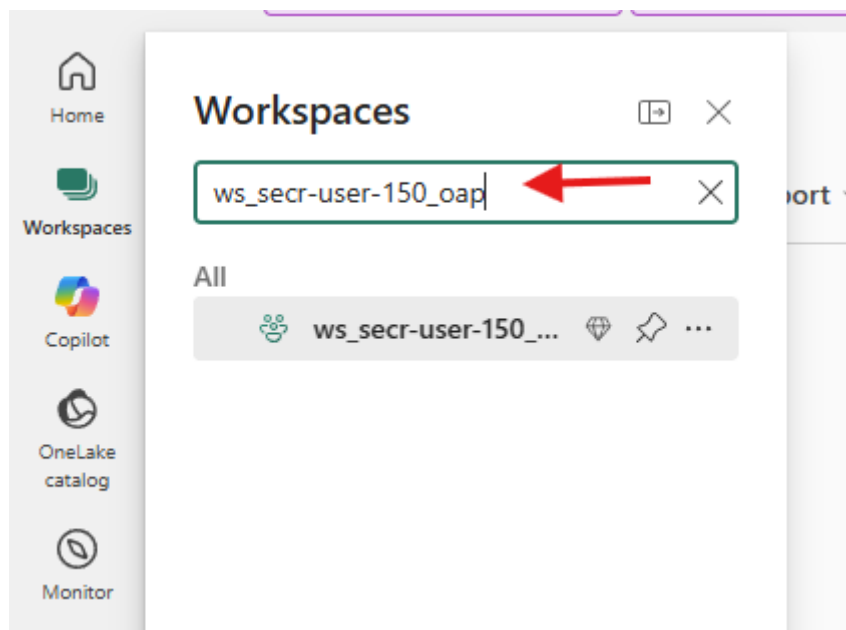
Once configured, you will then test outbound connectivity to write to an external Azure Storage account and another workspace in the same tenant.

For this exercise, you will use pre-configured workspace **ws_<user>_oap**.

Hands-on: Setup managed private endpoint and connect to ADLS g2 from OAP workspace

Step 1: Enable OAP and configure managed private endpoint (MPE) for ADLS.

1. Go to Fabric workspace and navigate to pre-provisioned workspace for OAP.



2. Navigate to **Workspace Settings > Outbound networking**.

The screenshot shows the 'Workspace settings' page in Azure Databricks. The left sidebar contains a list of settings categories: General, Workspace type, Azure connections, System storage, Git integration, OneLake, Workspace identity, Outbound networking (highlighted with a red box), Inbound networking, Encryption, Monitoring, Power BI, Delegated Settings, Data Engineering/Science, Data Factory, and Data Warehouse. The main content area is titled 'Network security' and includes a description of managed private endpoints. Below this is a table of 'Managed private endpoints' with columns for Name, Activation, and Approval. The table contains one entry: 'mpe-ws-secr-user-150-wspl' with 'Succeeded' activation and 'Approved' approval. At the bottom, the 'Outbound access protection' section is visible, showing two toggles: 'Block outbound public access' and 'Allow Git Integration', both currently set to 'Off'.

3. Enable outbound access protection for the workspace using the toggle. Once enabled, all outbound connections by supported Fabric workloads will be blocked by default.

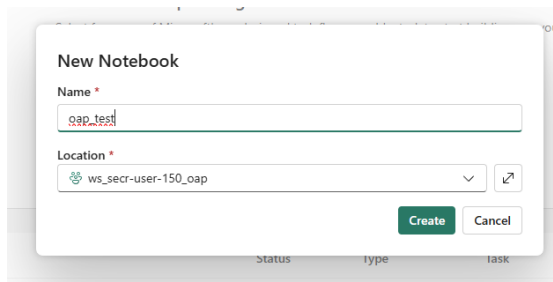
This is a close-up of the 'Outbound access protection' settings. It shows two toggle switches, both currently in the 'Off' position. The first toggle is 'Block outbound public access', which is described as blocking all outbound connections from the workspace and only allowing connections through private endpoints and connection rules. The second toggle is 'Allow Git Integration', which is described as allowing all Git features like commit and sync for the repository connected to the workspace. Both toggles are currently set to 'Off'.

4. Change this toggle to ON; when prompted, mark the checkbox and click on **Yes**.

The screenshot shows a confirmation dialog titled 'Block outbound access on this workspace?'. It contains a message explaining that blocking outbound public access is only available for a limited set of items and that creation of other items will be disabled. Below the message is a checkbox labeled 'I understand that certain artifacts cannot be created once this setting is enabled.', which is checked. At the bottom right are two buttons: 'Yes' and 'Cancel'.

5. Once OAP is enabled for the workspace, you can test writing to an ADLS account using a notebook without a managed private endpoint (MPE).

6. Create a New Item of type Notebook called “oap_test”



The screenshot shows a 'New Notebook' dialog box. It has a title bar 'New Notebook'. Below the title bar, there are two fields: 'Name' and 'Location'. The 'Name' field contains the text 'oap_test'. The 'Location' field contains a dropdown menu with the text 'ws_secr-user-150_oap'. To the right of the dropdown menu is a small icon of a document with an arrow. At the bottom right of the dialog box, there are two buttons: 'Create' and 'Cancel'.

New Notebook

Name *

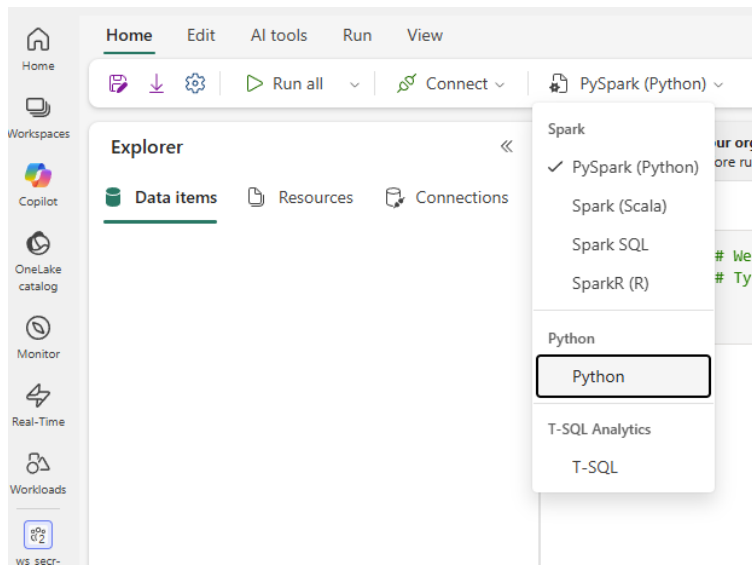
oap_test

Location *

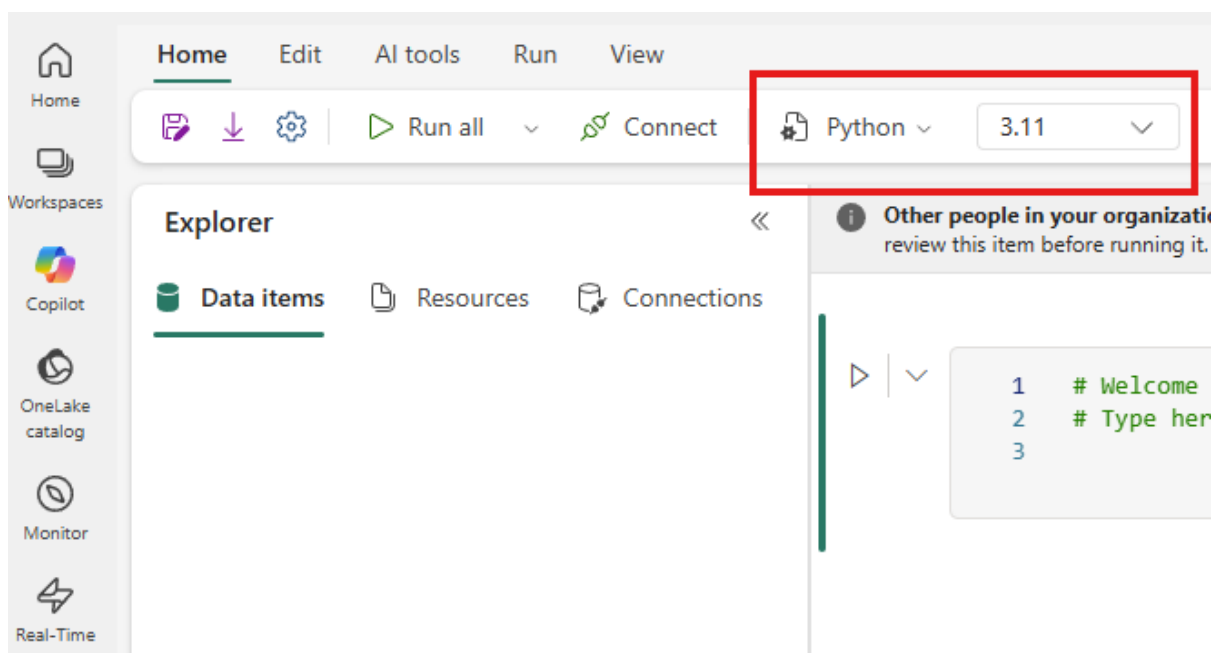
ws_secr-user-150_oap

Create Cancel

7. Change the notebook type from PySpark(Python) to Python



Python notebooks are generally faster to start up than PySpark notebooks.



8. Paste the below code snippet (replace the container with your username) in a new cell and run. This should fail as outbound is blocked from this workspace.

```
import notebookutils
```

```
container = "secre-user-150" #replace this with your username  
storageaccount = "adlsfabconsec"  
directory = "fabric_mpe_test"
```

```
test_message = "This is a sample created for testing MPE."
```

```

file_name = (notebookutils.runtime.context['userName']).lower().replace(
    ",", "") + ".txt"
adls_path =
f"abfss://{container}@{storageaccount}.dfs.core.windows.net/{directory}/{file
name}"

#This will write file to the ADLS path.

notebookutils.fs.put(adls_path, test_message, True)

```

9. With outbound connectivity blocked and absence of a managed private endpoint (MPE) to ADLS, the code will fail with the following error.

```

''' attempting connection to external endpoint
ServiceRequestError
(<urllib3.connection.HTTPSConnection object at 0x7b56f47854d0>, 'Connection to adlsfabconsec.blob.core.windows.net timed out.
(connect timeout=20)')

```

Show traceback 

10. You will now configure an MPE to external ADLS account. Navigate to **Workspace Settings** for the workspace ending with *_oap*.

11. Navigate to **Outbound networking** and then select **+Create** under **Managed private endpoints**.

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption
- Monitoring

Power BI ▾
Delegated Settings ▾
Data Engineering/Science ▾
Data Factory ▾
Data Warehouse ▾

Network security

Managed private endpoints let people securely connect to an Azure resource or Private Link service. [Learn more](#)

Managed private endpoints

Managed private endpoints are currently available for notebooks and Spark job definitions. [Learn more](#)

[+ Create](#) [Refresh](#) [Delete](#)

☐	Name ↓	Activation	Approval
☐	mpe-ws-secr-user-150-wspl	Succeeded	Approved

Outbound access protection

Block outbound public access ☒ On

Block all outbound connections from workspace and only allow connections through private end points and connection rules. This feature is only available for limited items. Creation of other items will be disabled once this feature is turned on. Please wait for 15 mins for the setting change to take effect. [Learn more](#)

Allow Git Integration ☐ Off

Use all Git features, like commit and sync, for the repository connected to this workspace. If this is off, Git integration will be blocked.

Data connection rules (preview)

[+ Add new cloud connection rule](#) [Save](#)

Cloud connection policies

> All other connection kinds ☐ Blocked

Gateway connection policies

> Virtual network and On-premises data gateways ☐ Blocked

12. Create a new Managed Private Endpoint and fill out the details for the managed private endpoint. You can use the following values for configuring MPE.

- **Managed private endpoint name:** **mpe-username-dfs** (example: mpe-secr-user-150-dfs)
- **Resource identifier:** This is the resource ID from Azure. In this workshop we will be sharing a single ADLS account so you can use the following resource ID value.

```
/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec
```

- **Target sub-resource:** Azure Data Lake Storage Gen2

- **Request message:** This is text field but for this workshop, please put down your username-dfs here. This will make it easier for you to identify your request in Azure portal. **(Example: secr-user-150-dfs)**

The screenshot shows a modal dialog titled "Create managed private endpoint" with a close button (X) in the top right corner. Below the title is a descriptive sentence: "Create a managed private endpoint to securely connect this Fabric workspace to your datasources." The dialog contains four input fields: "Managed private endpoint name" with the value "mpe-secr-user-150-dfs", "Resource identifier" with the value "/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/p", "Target sub-resource" with a dropdown menu showing "Azure Data Lake Storage Gen2", and "Request message" with the value "secr-user-150-dfs". At the bottom right, there are two buttons: a green "Create" button and a white "Cancel" button.

Once details have been entered, click **Create**.

13. You will see that the MPE is provisioning, wait for 5 minutes, in the meantime, proceed to next step.

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption

Network security

Managed private endpoints let people securely connect to an Azure resource or Private Link service. [Learn more](#)

Managed private endpoints

Managed private endpoints are currently available for notebooks and Spark job definitions. [Learn more](#)

[+ Create](#) [Refresh](#) [Delete](#)

☐	Name ↓	Activation	Approval
☐	mpe-secr-user-150-dfs	Provisioning	-
☐	mpe-ws-secr-user-150-wspl	Succeeded	Approved

14. For **Azure Storage**, you must create separate managed private endpoints for Blob (*.blob.core.windows.net) and Data Lake storage (*.dfs.core.windows.net) service. You can see full list of endpoints associated with Azure Storage by navigating to **Azure portal > Azure Storage > Settings > Endpoints**.

The screenshot displays the 'Endpoints' page in the Azure portal for a storage account named 'adlsfabconsec'. The left-hand navigation pane shows the 'Settings' menu expanded, with 'Endpoints' selected. The main content area lists the provisioning state, creation time, and resource ID for several services. The 'Blob service' and 'Data Lake Storage' sections are highlighted with red boxes. The 'Blob service' section shows the endpoint URL 'https://adlsfabconsec.blob.core.windows.net/'. The 'Data Lake Storage' section shows the resource ID '/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro'.

Service	Resource ID	Endpoint URL
Blob service	/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro	https://adlsfabconsec.blob.core.windows.net/
File service	/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro	https://adlsfabconsec.file.core.windows.net/
Queue service	/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro	https://adlsfabconsec.queue.core.windows.net/
Table service	/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro	https://adlsfabconsec.table.core.windows.net/
Data Lake Storage	/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/pro	

You will observe separate endpoints OAP is a network-layer feature. For each external endpoint (which is the case here with Azure Storage), one must configure separate MPE for Blob and Data Lake storage endpoints. These are commonly used by Fabric for accessing data from Azure Storage. You will do that in the following step.

15. Create a new managed PE in addition to the one created in previous step – but this time you will choose **Azure Blob Storage for Target Sub-resource**:

MPE Name: mpe-username-blob (Example: mpe-secr-user-150-blob)

Resource ID: /subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec

Target sub-resource: **Azure Blob Storage**

Request Message: username-blob (Example: secr-user-150-blob)

Create managed private endpoint

Create a managed private endpoint to securely connect this Fabric workspace to your datasources.

Managed private endpoint name ⓘ

mpe-secr-user-150-blob

Resource identifier ⓘ

/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/p

Target sub-resource

Azure Blob Storage

Request message ⓘ

secr-user-150-blob

Create Cancel

Cloud connection policies

Use the following values for configuring this second MPE:

- **Managed private endpoint name:** This is a text field so you can enter any name. For this workshop, we recommend including your username so that it will be easier to identify the endpoint provisioned by you.
mpe_<username>.
Resource identifier: /subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec
- **Target sub-resource:** Select **Azure Blob Storage**.

Create managed private endpoint

×

Create a managed private endpoint to securely connect this Fabric workspace to your datasources.

Managed private endpoint name ⓘ

blobmpe_secr-test-001

Resource identifier ⓘ

/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/p

Target sub-resource

Azure Blob Storage

Request message ⓘ

This is a MPE request from user secr-test-001.

CreateCancel

Select **Create** once all details have been entered.

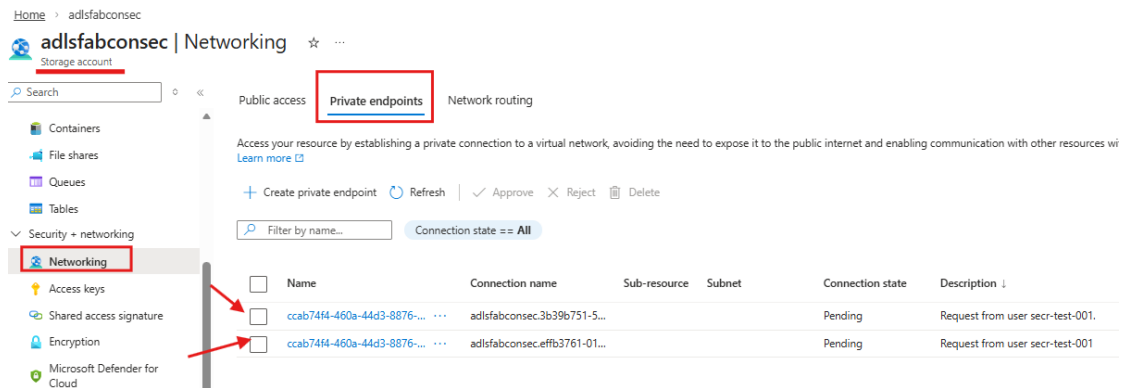
16. You should see that the **Activation** status change to **provisioning**.

+ Create Refresh Delete			
☐	Name ↓	Activation	Approval
☐	 mpe_secr-test-001	 Provisioning	 -

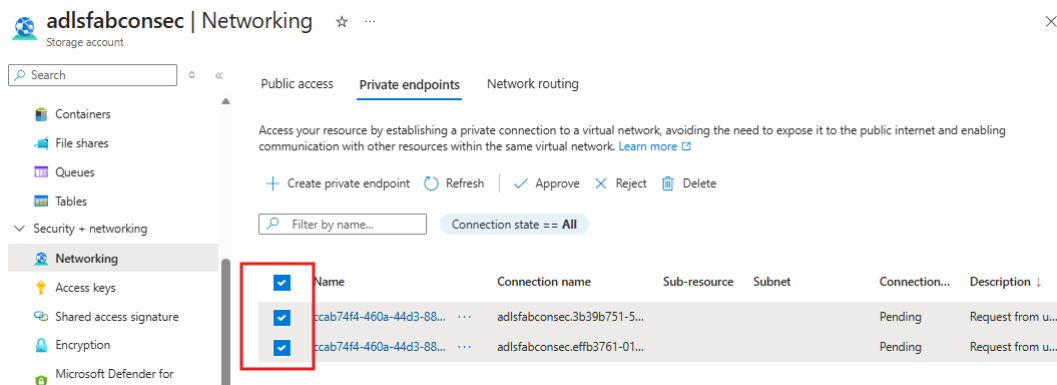
17. After 2-3 minutes, the status should change to **Succeeded**. At this stage, the resource administrator (in this case, ADLS) is ready to review and approve MPE request.

+ Create Refresh Delete			
☐	Name ↓	Activation	Approval
☐	 mpe_secr-test-001	 Succeeded	 Pending

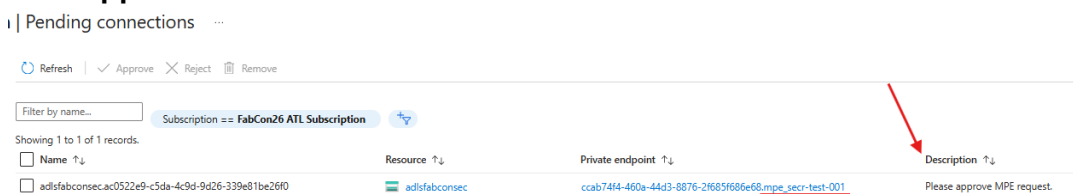
18. Switch to **Azure portal** and login using username and password provisioned for you for this workshop. Navigate to **Storage account > Networking > Private endpoints**. Here, you should see 2 requests for private endpoints in **Pending** state.



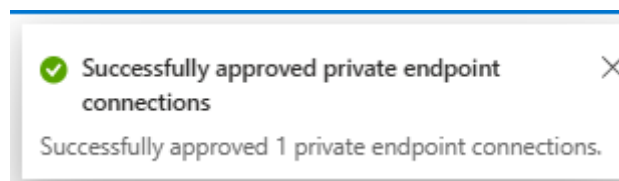
19. Select the two requests for your user/workspace and then select **Approve**.



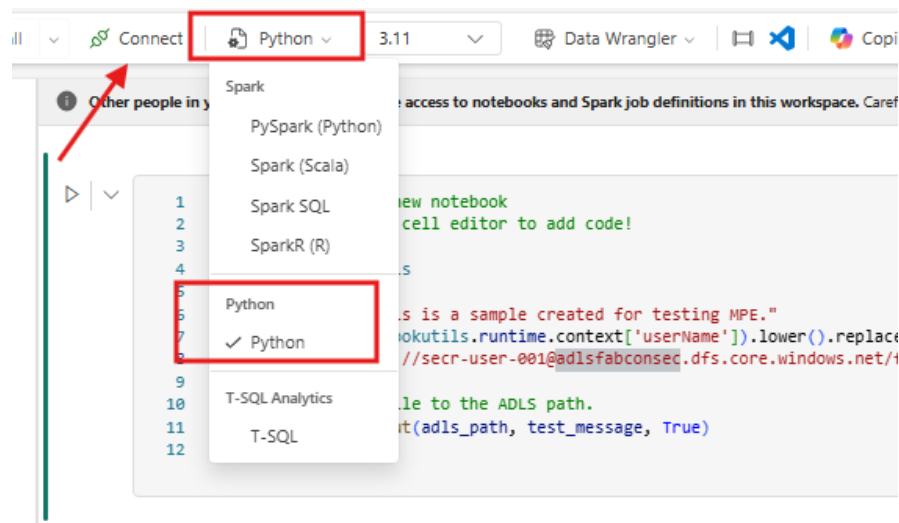
20. Since you are sharing this environment with other users, **Pending Connections** will show all the connection requests submitted by other users. Please identify the connection requested by you and approve it. In this case, we used **adlsmpe_sec-test-001** as the name. Select the request which was submitted by you and click **Approve**.



21. Once approved, you should see a pop-up on the top-right corner inside Azure portal.



25. Please ensure that you are using **Python**. This has smaller footprint compared to Spark. Once you have switched to Python, select **Connect**.



It takes a few minutes for compute to provision and start for OAP enabled workspaces. You should see status switch from **Connecting ...** to **Connected** when it's ready to go.



26. In the notebook cell, paste this code. Ensure that you replace variable values prior to running the code. This sample code will create a file on ADLS. If MPE for ADLS has been configured correctly, you should be able to write a file out to ADLS.

```
import notebookutils

container = "secre-user-150" #replace this with your username
storageaccount = "adlsfabconsec"
directory = "fabric_mpe_test"

test_message = "This is a sample created for testing MPE."
file_name = (notebookutils.runtime.context['userName']).lower().replace(" ", "") + ".txt"
adls_path =
f"abfss://{container}@{storageaccount}.dfs.core.windows.net/{directory}/{file_name}"

#This will write file to the ADLS path.
```

```
notebookutils.fs.put(adls_path, test_message, True)
```

Replace the parameters with the following values:

- **<container>**: Container pre-created for this workshop. This should be same as your username.
- **<storageaccount>**: *adlsfabconsec*
- **<directory>**: *abric_mpe_test*

The code should print out **True** once it has completed running. You can confirm that the file has been written by switching to **Azure Portal** and navigating to the storage account.

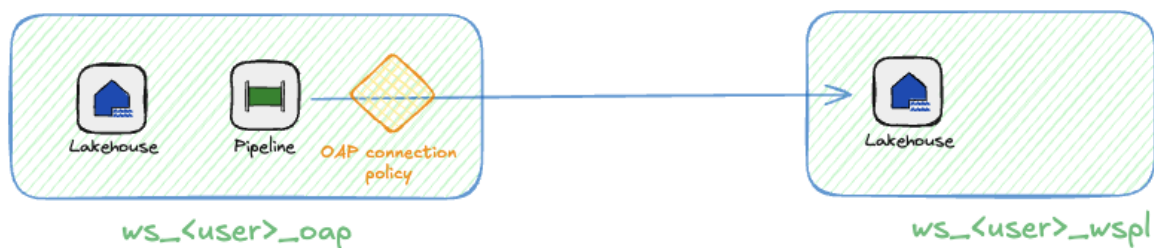
Checkpoint: At this stage, you have:

- Enabled OAP on your workspace ending with *_oap*.
- Created a managed private endpoint for ADLS and approved it via Azure Portal.
- Tested outbound connectivity to ADLS with and without MPE

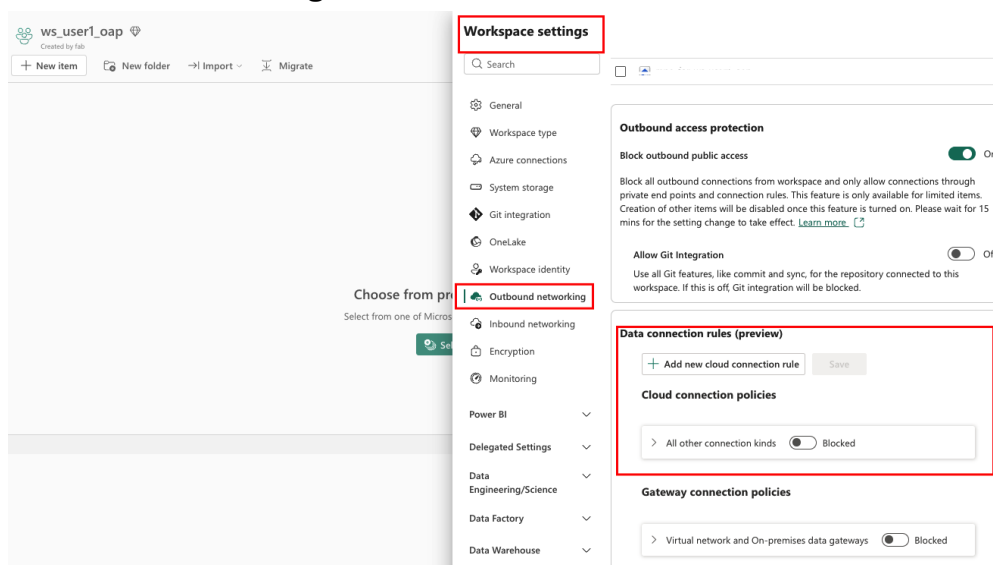
Step 2: Setup connection rules for Data Factory workloads

When outbound access protection is enabled for a workspace, all outbound connections are blocked by default. For Data Factory workloads, you can permit access to external data sources by configuring data connection rules.

In this exercise, we will be using **cloud connections** to enable outbound access for Data Factory workloads (Pipelines, Dataflows gen2). You will setup policy to allow pipeline from OAP enabled workspace to write data out to a lakehouse in **ws_<user>_public** workspace.



27. Navigate to workspace **ws_<user>_oap** and select **Workspace Settings > Outbound networking. B**



28. What you are doing here is that you are allowing outbound to lakehouse in workspace **ws_<user>_wspl**. Enter the following details.

- **Connection kind:** Lakehouse
- **Workspace exception:** *Enter GUID of ws_<user>wspl workspace*

Data connection rules (preview)

New rule

Connection kind

Lakehouse

Workspace exception

ws_secr-user-150_public (412cc558-9b4b-49f8-b31c-af5...)

Add

29. If configured correctly, you should see workspace ws_<user>_wspl under **Connection Policies**. Select **Save** to apply these settings.

Data connection rules (preview)

+ Add new cloud connection rule **Save**

Cloud connection policies

▼ Lakehouse ☐ Blocked

Exceptions: Allowed workspaces: 1 (Current workspace is always allowed)

Workspaces

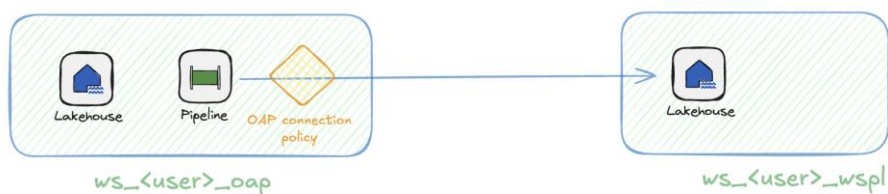
ws_secr-user-150_public (412cc558-9b4b-49f8-b31c-af529838f938)

> All other connection kinds ☐ Blocked

Gateway connection policies

> Virtual network and On-premises data gateways ☐ Blocked

30. After the connection policy is saved, create a pipeline that **reads** from workspace **ws_<user>_oap** and writes the output to a lakehouse table in external workspace **ws_<user>_wspl**.



31. Create a new **Pipeline** in OAP workspace **ws_<user>_oap** assigned to you. Please replace **<user>** with the user ID assigned to you for the workshop.

New Pipeline

Name *

pipe_from_ws_oap_to_external_lh

Location *

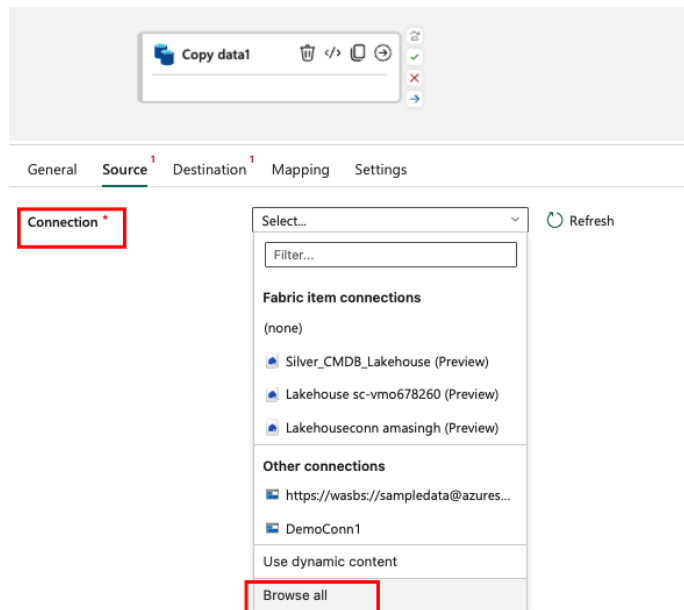
ws_user1_wspl

☒ Open artifact when creation succeeded ⓘ

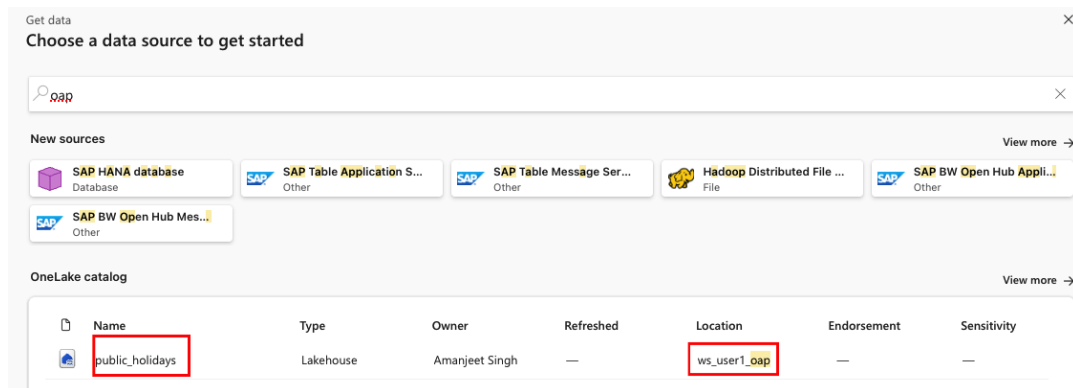
Create

Cancel

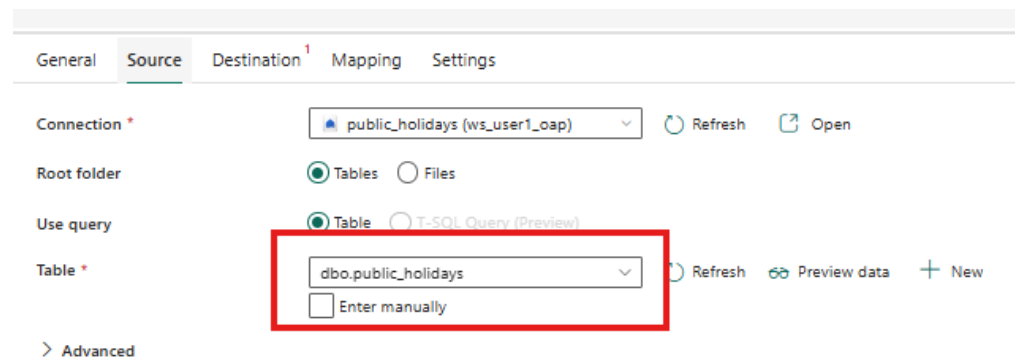
32. In Pipeline authoring experience, select the table created in the OAP workspace **ws_<user>_oap** as the source and the lakehouse **ws_<user>_public** as the target.



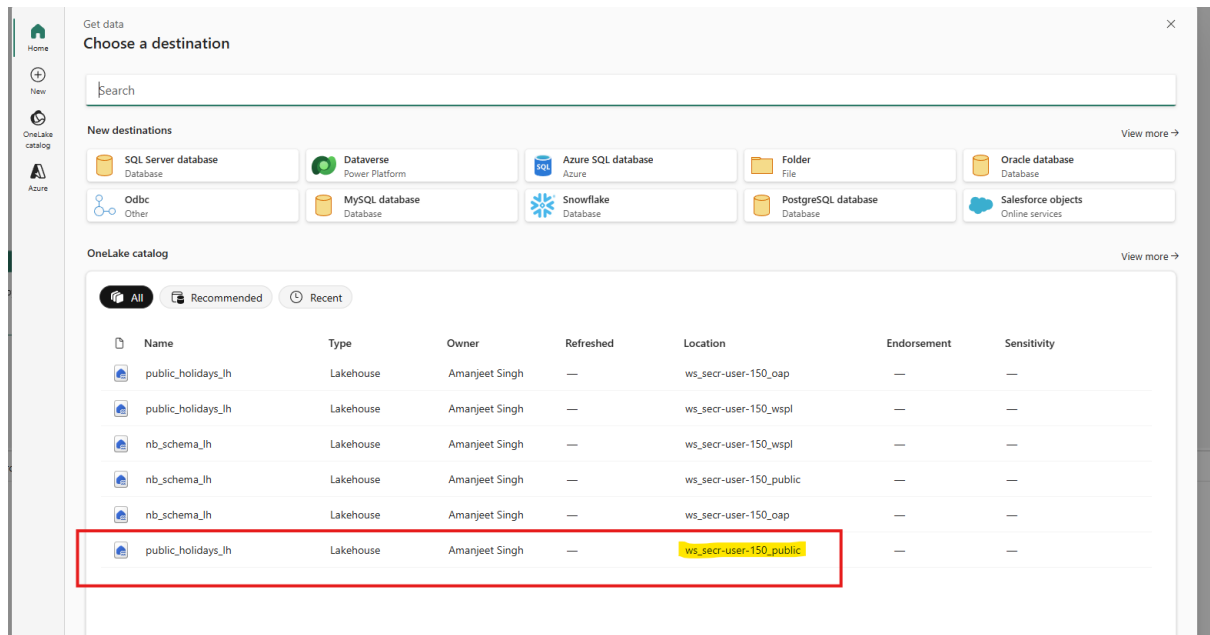
33. Select the source **lakehouse**.



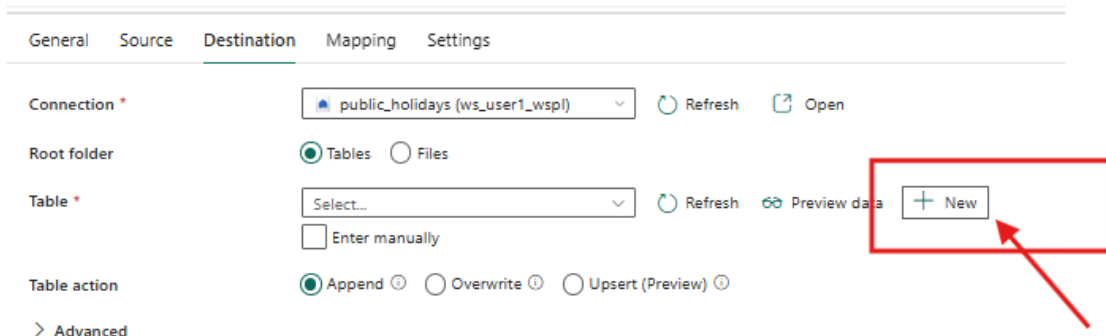
34. Select table **public_holidays**.



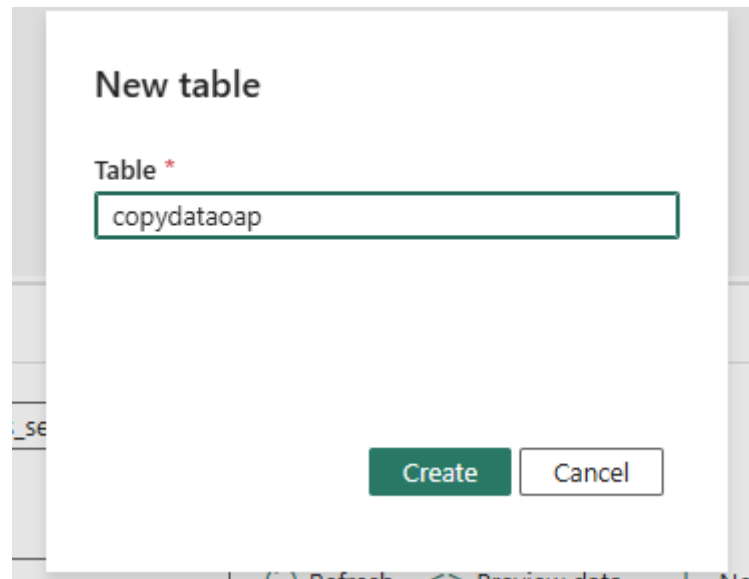
35. Select **destination** which is the pre-provisioned lakehouse in **ws_<user>_public** workspace.



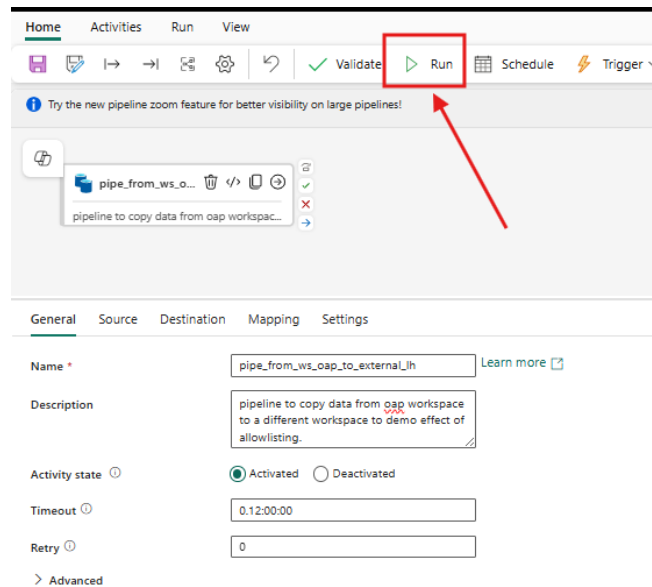
36. Select **+New** table. This table is the destination for data copied from the restricted workspace **ws_<user>_oap**, where outbound data movement is blocked. Because the target lakehouse is allowlisted via **Connection Policy**, Fabric permits writing data to this external lakehouse.



37. Enter the details of the target table and select **Create**. Please ensure that you enter appropriate details for the target table. In this example, we used schema **dbo** and table name **copydataoap**.



38. Select **Run** to execute the pipeline.



39. Monitor the pipeline activity in the **Output** pane at the bottom of the session.
When execution completes, the **Activity Status** transitions from **Queued** → **In Progress** → **Succeeded**.

HomeActivitiesRunView

Try the new pipeline zoom feature for better visibility on large pipelines!

pipe_from_ws_oap_to_exter...
pipeline to copy data from oap workspac...

ParametersVariablesSettingsOutputLibrary variables

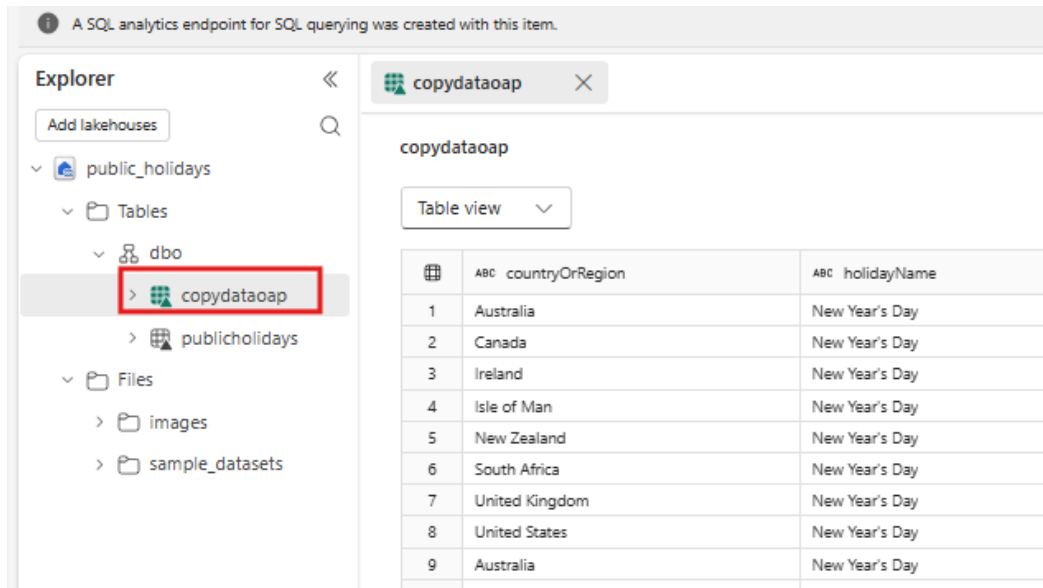
Pipeline run ID 11384048-8429-4da4-8f1a-bd86a9d20d9e Pipeline status Succeeded

Filter by keyword

Showing 1 - 1 items

Activity name	Activity status	Run start	Duration
pipe_from_ws_oap_to_external_lh	Succeeded	2/28/2026, 3:01:42 PM	18s

40. Navigate to workspace **ws_<user>_public** and select the lakehouse. You should now see the new table which you created by initiating a pipeline in OAP restricted workspace (**ws_<user>_oap**).



The screenshot shows the Databricks workspace interface. On the left, the 'Explorer' sidebar is visible, showing a tree view of the workspace. The 'public_holidays' lakehouse is expanded, and the 'copydataoap' table is highlighted with a red box. The main panel on the right displays the 'copydataoap' table in 'Table view' mode. The table has two columns: 'countryOrRegion' and 'holidayName'. The data shows 'New Year's Day' for various countries including Australia, Canada, Ireland, Isle of Man, New Zealand, South Africa, United Kingdom, United States, and Australia.

	countryOrRegion	holidayName
1	Australia	New Year's Day
2	Canada	New Year's Day
3	Ireland	New Year's Day
4	Isle of Man	New Year's Day
5	New Zealand	New Year's Day
6	South Africa	New Year's Day
7	United Kingdom	New Year's Day
8	United States	New Year's Day
9	Australia	New Year's Day

This completes **Connection Policy** module which is part of OAP.

You learnt how to configure an allowlist to enable outbound connectivity from a workspace with OAP enabled, and how to use a pipeline activity to copy data to an external workspace or lakehouse.

Summary

In this module, you examined how Microsoft Fabric enforces outbound access controls at the workspace level using Outbound Access Protection (OAP). You enabled OAP to block all outbound connectivity by default and configured managed private endpoints to explicitly allow secure, private access to external Azure services.

You deployed a **managed private endpoint** from a Fabric workspace to Azure Data Lake Storage Gen2, reviewed the approval workflow on the resource owner side, and validated end-to-end connectivity using a Spark notebook.

This was followed by setting up a **Connection Policy** for Data Factory workloads such as Pipelines to communicate with external services and applications including other workspaces. You created a policy to allow connectivity to a lakehouse in second workspace so that data can be copied across successfully.

Managed PE and Connection Policy features ensure that data egress occurs only through explicitly sanctioned private connections.

This concludes Module 2 – Outbound Protection.